Accelerated Math II/III Name_____

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HW #12 - Fundamental Theorem of Algebra

Date_____ Period____

State the number of complex zeros, the possible number of real and imaginary zeros, and the possible rational zeros for each function.

1) $f(x) = x^5 - 3x^4 + 8x^3 - 24x^2 + 15x - 45$

2) $f(x) = x^6 - 4x^4 - 4x^2 + 16$

3) $f(x) = x^2 - 8x + 16$

4) $f(x) = x^3 - x^2 - 5x - 3$

Find all zeros. One zero has been given.

5) $f(x) = 2x^4 - 5x^3 - 4x^2 + 3x;\ 3$

6) $f(x) = 3x^4 - 14x^3 + 27x^2 - 36x;\ 3$

7) $f(x) = 10x^4 + 45x^3 + 41x^2 - 18x - 18;\ -\frac{3}{2}$

8) $f(x) = 10x^4 + 48x^3 + 45x^2 - 11x;\ \frac{1}{5}$

State the possible rational zeros for each function. Then find all zeros.

9) $f(x) = x^6 - 4x^4 - x^2 + 4$

10) $f(x) = x^5 - 3x^4 + 4x^3 - 12x^2 + 3x - 9$

11) $f(x) = x^2 - 14x + 13$

12) $f(x) = x^6 - 126x^3 + 125$

Use the information provided to write the vertex form equation of each parabola.

13) Vertex: $(8, 1)$, Focus: $\left(8, \frac{7}{8}\right)$

14) Vertex: $(0, 4)$, Focus: $\left(-\frac{1}{8}, 4\right)$

15) Focus: $(8, 2)$, Directrix: $y = -8$

16) Focus: $\left(5, \frac{7}{8}\right)$, Directrix: $y = \frac{9}{8}$

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Answers to HW #12 - Fundamental Theorem of Algebra

1) # of complex zeros: 5
Possible # of real zeros: 5, 3, or 1
Possible # of imaginary zeros: 4, 2, or 0
Possible rational zeros:
$\pm 1, \pm 3, \pm 5, \pm 9, \pm 15, \pm 45$

2) # of complex zeros: 6
Possible # of real zeros: 6, 4, 2, or 0
Possible # of imaginary zeros: 6, 4, 2, or 0
Possible rational zeros: $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16$

3) # of complex zeros: 2
Possible # of real zeros: 2 or 0
Possible # of imaginary zeros: 2 or 0
Possible rational zeros: $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16$

4) # of complex zeros: 3
Possible # of real zeros: 3 or 1
Possible # of imaginary zeros: 2 or 0
Possible rational zeros: $\pm 1, \pm 3$

5) $\left\{0, \frac{1}{2}, -1, 3\right\}$

6) $\left\{0, \frac{5+i\sqrt{119}}{6}, \frac{5-i\sqrt{119}}{6}, 3\right\}$

7) $\left\{-3, \frac{\sqrt{10}}{5}, -\frac{\sqrt{10}}{5}, -\frac{3}{2}\right\}$

8) $\left\{0, \frac{-5+\sqrt{3}}{2}, \frac{-5-\sqrt{3}}{2}, \frac{1}{5}\right\}$

9) Possible rational zeros: $\pm 1, \pm 2, \pm 4$
Zeros: $\{2, -2, 1, -1, i, -i\}$

10) Possible rational zeros: $\pm 1, \pm 3, \pm 9$
Zeros: $\{3, i, -i, i\sqrt{3}, -i\sqrt{3}\}$

11) Possible rational zeros: $\pm 1, \pm 13$
Zeros: $\{1, 13\}$

12) Possible rational zeros: $\pm 1, \pm 5, \pm 25, \pm 125$
Zeros: $\left\{5, \frac{-5+5i\sqrt{3}}{2}, \frac{-5-5i\sqrt{3}}{2}, 1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}\right\}$

13) $y = -2(x-8)^2 + 1$

14) $x = -2(y-4)^2$

15) $y = \frac{1}{20}(x-8)^2 - 3$

16) $y = -2(x-5)^2 + 1$

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